AT-WORST-NODAL RELATIVE DIMENSION 1 DESCENDS ALONG DIRECTED LIMITS OF AFFINE TRANSITION MORPHISMS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the syntomic descent, the relative-dimension decomposition, the constructible containment descent, and the unramifiedness descent are all retained verbatim.

Editorial note. The label lemma-descend-nodal-family names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Descent of the syntomic property. Limits, Lemma limits-lemma-descend-syntomic. Produces an index i at which the base-changed morphism f_i is syntomic.
- Syntomic relative dimension and constructible containment. Morphisms, Lemmas morphisms-lemma-syntomic-relative-dimension and morphisms-lemma-dimension-fibre-after-base-change; Limits, Lemma limits-lemma-limit-contained-in-constructible. Decomposes X_i into open-and-closed relative-dimension pieces, shows the limit X maps into the dimension-1 piece, and descends that containment to some finite level.
- Descent of unramifiedness. Limits, Lemma limits-lemma-descend-unramified. Promotes the unramifiedness of Sing(f) over Y, known after base change to Y, back to some finite level i'.

Source reference. Original lemma 0C3L: sources/stacks-chow/tags.txt, source lines 15196–15196. Prerequisite pointer invoked at proof line(s) 5113; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0C4W: sources/stacks-chow/tags.txt, source lines 15241–15241. Prerequisite pointer invoked at proof line(s) 5130; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 05F4: sources/stacks-chow/tags.txt, source lines 7013–7013. Prerequisite pointer invoked at proof line(s) 5124; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original lemma 02FY: sources/stacks-chow/tags.txt, source lines 3209–3209. Prerequisite pointer invoked at proof line(s) 5120; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02K1: sources/stacks-chow/tags.txt, source lines 3356–3356. Prerequisite pointer invoked at proof line(s) 5118; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $S = \lim S_i$ be a limit of a directed system of schemes with affine transition morphisms. Let $0 \in I$ and let $f_0 : X_0 \to Y_0$ be a morphism of schemes over S_0. Assume S_0, X_0, Y_0 are quasi-compact and quasi-separated. Let $f_i : X_i \to Y_i$ be the base change of f_0 to S_i and let $f : X \to Y$ be the base change of f_0 to S. If*

(1) *f is at-worst-nodal of relative dimension 1, and*
(2) *f_0 is locally of finite presentation,*

then there exists an $i \geq 0$ such that f_i is at-worst-nodal of relative dimension 1.

Proof. By Limits, Lemma 0C3L there exists an i such that f_i is syntomic. Then $X_i = \coprod_{d \geq 0} X_{i,d}$ is a disjoint union of open and closed subschemes such that $X_{i,d} \to Y_i$ has relative dimension d, see Morphisms, Lemma 02K1. Because of the behaviour of dimensions of fibres under base change given in Morphisms, Lemma 02FY we see that $X \to X_i$ maps into $X_{i,1}$. Then there exists an $i' \geq i$ such that $X_{i'} \to X_i$ maps into $X_{i,1}$, see Limits, Lemma 05F4. Thus $f_{i'} : X_{i'} \to Y_{i'}$ is syntomic of relative dimension 1 (by Morphisms, Lemma 02FY again). Consider the morphism $\mathrm{Sing}(f_{i'}) \to Y_{i'}$. We know that the base change to Y is an unramified morphism. Hence by Limits, Lemma 0C4W we see that after increasing i' the morphism $\mathrm{Sing}(f_{i'}) \to Y_{i'}$ becomes unramified. This finishes the proof. □